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Clinical Investigation: Head and Neck Cancer

Osteoradionecrosis and Radiation Dose to the Mandible in Patients With Oropharyngeal Cancer

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Summary

The records of 402 oropharyngeal cancer patients treated with definitive radiation therapy were reviewed for the occurrence of osteoradionecrosis (ORN). In 30 patients (7.5%), ORN developed, including 6 patients with grade 4 ORN that required major surgery. The median time to develop ORN was 8 months. Compared with ORN-free patients, patients with ORN had significantly larger volumes of mandible receiving 50 Gy (V50) and 60 Gy (V60), suggesting that reducing V50 may decrease ORN risk.

Purpose: To determine the association between radiation doses delivered to the mandible and the occurrence of osteoradionecrosis (ORN).

Methods and Materials: We reviewed the records of 402 oropharyngeal cancer patients with stage T1 or T2 disease treated with definitive radiation between January 2000 and October 2008 for the occurrence of ORN. Demographic and treatment variables were compared between patients with ORN and those without. To examine the dosimetric relationship further, a nested case-control comparison was performed. One to 2 ORN-free patients were selected to match each ORN patient by age, sex, radiation type, treatment year, and cancer subsite. Detailed radiation treatment plans for the ORN cases and matched controls were reviewed. Mann-Whitney test and conditional logistic regression were used to compare relative volumes of the mandible exposed to doses ranging from 10 Gy-60 Gy in 10-Gy increments.

Results: In 30 patients (7.5%), ORN developed during a median follow-up time of 31 months, including 6 patients with grade 4 ORN that required major surgery. The median time to develop ORN was 8 months (range, 0-71 months). Detailed radiation treatment plans were available for 25 of the 30 ORN patients and 40 matched ORN-free patients. In the matched case-control analysis, there was a statistically significant difference between the volumes of mandible in the 2 groups receiving doses between 50 Gy (V50) and 60 Gy (V60). The most notable difference was seen at V50, with a P value of .02 in the multivariate model after adjustment for the matching variables and dental status (dentate or with extraction).

Conclusions: V50 and V60 saw the most significant differences between the ORN group and the comparison group. Minimizing the percent mandibular volume exposed to 50 Gy may reduce ORN risk. © 2013 Elsevier Inc.

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Introduction

Osteoradionecrosis (ORN) is a potentially debilitating complication of radiation therapy for patients with head-and-neck cancers. Possible risk factors include age, sex, medical comorbidities, primary tumor site and stage, tumor location in relation to the mandible, dentition status, types of treatment and technique (external beam radiation, brachytherapy, surgery, chemotherapy, or combination therapy), radiation dose, dental extraction before and after radiation or surgery, smoking, and alcohol (1-3).

In a study of head-and-neck cancer patients treated with parotid-sparing intensity modulated radiation therapy (IMRT) by Ben-David et al, prophylactic dental care and the use of IMRT resulted in no case of grade $\geq$2 ORN (1). Similar results were supported by other studies (4-6). Currently, the most commonly recommended dose constraint to the mandible is a maximum of 70 Gy (3).

Given the relatively low incidence of ORN, the number of ORN cases included in published literature is small. There is no multivariate analysis comparing mandibular radiation doses between ORN and ORN-free patients, as far as we are aware. This study examined the association between radiation doses to the mandible and ORN in patients treated with 3-dimensional (3D) conformal radiation therapy or IMRT. We compared mandibular dose distributions in ORN and ORN-free patients to establish a clinically relevant mandibular dose constraint. We also analyzed multiple possible risk factors associated with ORN.

Methods and Materials

Patient selection

This study included 402 patients with newly diagnosed oropharyngeal cancer, having clinically staged T1 or T2 disease and treated between January 2000 and October 2008. They were part of a prospective molecular epidemiologic study of genetic susceptibility to head-and-neck cancers. The Institutional Review Board at our institution approved this study, and informed consent was obtained for each patient. We selected only those with clinical stage T1-T2 disease and those treated with definitive radiation therapy (with or without chemotherapy) to achieve a relatively homogenous cohort. We reviewed the records of the 402 patients to obtain demographic data, staging, treatment modality, dental records, ORN occurrence, and follow-up details.

Dental evaluation

All patients received a comprehensive dental evaluation by a dental oncologist before radiation therapy. Our dental oncologists used a set of well-defined guidelines/protocols to guide their preradiation evaluation and to minimize individual variation between dentists. Patients with poor dentition received preradiation dental extraction, with close attention paid to the posterior mandible (areas where molars and premolars are present). Each patient was given fluoride trays for dental hygiene practice. Dental records were reviewed, and patients were divided into 3 categories: (*1*) dentate in the posterior mandible without extraction, (*2*) dental extraction performed in the posterior mandible or elsewhere, and (*3*) edentulous in the posterior mandible at baseline.

Radiation treatment

Patients were treated with 3D conformal or IMRT. The 3D treatment typically used a 3-field arrangement with subsequent cone-down technique. IMRT used a multileaf collimator in a step-and-shoot, multiple static beam arrangement (7). Treatments for our study group were planned using the CORVUS treatment planning system (CORVUS version 4.0; Nomos Corporation, Pittsburgh, PA) or a Pinnacle system (version 6.2b or later, Philips Medical Systems). Radiation was delivered via 6-MV photons generated by a Varian linear accelerator (Varian Medical Systems, Palo Alto, CA).

Definition of ORN

The ORN was graded as follows: grade 1, minimal bone exposure with conservative management only; grade 2, minor debridement received; grade 3, hyperbaric oxygen needed; grade 4, major surgery required.

Patient follow-up

After radiation therapy, patients were followed up by a head-and-neck surgeon and a radiation oncologist, typically every 3 months in the first year, every 4-6 months in the following 2 years, then annually thereafter. Patients also returned for dental clinic visits for assessment of dentition and signs of ORN. Minor procedures and debridement were carried out during office visits if necessary. Patients who elected to be followed up by their local physicians were censored after their last visit at our institution.

Statistical analysis

For comparisons of basic demographic variables, clinical stage, and other general treatment-related information, all 402 patients were included. For evaluation of radiation doses to the mandible, we elected to do a matched, nested case-control analysis instead of reviewing the radiation plans for all 402 patients. At least 1 ORN-free patient was matched to each of the 25 ORN patients by age (within 10 years), sex, radiation type (3D or IMRT), treatment period (within 6 months), and cancer subsite. A second patient was selected if more than 1 ORN-free patient fulfilled the matching criteria. Detailed radiation treatment plans for the ORN cases and matched controls were reviewed. The mandible was contoured by 1 investigator for each plan, and relative dose-volume histograms were computed.

Statistical analyses were performed using SAS software (SAS Institute Inc, Cary, NC). A χ^2 test or Fisher's test was used to compare distributions of categoric variables, and Wilcoxon rank-sum test was used to compare the distributions of continuous variables. The Mann-Whitney statistical test was used to compare the ORN and matched ORN-free groups in the relative volumes of the mandible exposed to doses ranging from 10 Gy-60 Gy in 10-Gy increments. Conditional logistic regression was used to compare relative volumes of the mandible exposed to doses ranging from 10 Gy-60 Gy in 10-Gy increments while adjusting for the matching variables and dental status.

Results

Characteristics of all patients

The median follow-up time was 31 months (range, 1-120 months); ORN developed in 30 of the 402 patients (7.5%). The median time to develop ORN was 8 months (range, 0-71 months). Ten of the 30 ORN patients had exposed bone and received conservative management (grade 1), 9 had minor debridement (sequestrectomy, grade 2), 5 required hyperbaric oxygen (grade 3), and 6 patients had grade 4 ORN that required major surgery (including mandibulectomy or hemimandibulectomy, removal of necrotic bone from the mandible, laryngopharynectomy, multiple dental extractions and 4-quadrant alveoloplasty, and curettage of abscesses).

Osteoradionecrosis was more frequent among patients who smoked and consumed alcohol ($P=.005$ and .05, respectively) (Table 1). We observed a higher incidence of ORN among patients with larger tumors (stage T2 vs T1, $P=.05$). ORN was less common among patients treated with IMRT and to lower doses ($P=.07$). The 2 groups were similar in N category and overall group staging. There were no significant differences in the proportion of patients having surgery or in the proportion receiving chemotherapy.

Nested case-control analysis of ORN patients and selected ORN-free control individuals

Detailed radiation treatment plans were available for 25 of the 30 ORN patients (Table 2). The ORN patients and ORN-free control individuals also had a similar T category distribution ($P=1.0$) and radiation prescribed dose to the primary disease ($P=.43$). Significant differences were still present between the 2 groups of patients in their smoking history and alcohol consumption. The ORN patients more commonly were edentulous or required dental extraction compared with non-ORN patients. None of the ORN-free patients had pretreatment dental extraction, and the majority (38 patients, 93%) had intact teeth in the posterior mandible. Ten ORN patients (42%) were dentate in the mandible, and 9 patients (38%) had pretreatment dental extraction. A higher proportion of ORN patients were edentulous in the posterior mandible (5 patients, 21%) compared with ORN-free control individuals (3 patients, 7%). The difference in preradiation dental treatment between the 2 groups was statistically significant ($P<.0001$).

Dosimetric comparison and nested case-control analysis of 25 ORN patients and 40 selected ORN-free control individuals

As illustrated in the Fig. and in Table 3, there were statistically significant differences between the ORN and ORN-free groups in the percent volumes of the mandible receiving doses between 40 Gy (V40) and 60 Gy (V60) according to either a nonparametric test or conditional logistic regression. The difference at V50 was most notable (40.5% vs 30.8%, $P=.004$ from nonparametric test). When ORN cases were compared with ORN-free controls, univariate conditional logistic regression demonstrated a significantly increased risk of ORN with increased V40 (odds ratio [OR] = 1.05; 95% confidence interval [CI] = 1.01-1.10; $P=.01$).

Table 1 Characteristics of the study cohort

Characteristic	ORN, n (%)	ORN-free, n (%)	*P*
Median age at diagnosis, y (range)	52 (35-80)	53 (28-81)	.94
Sex			
Female	2 (6.7%)	42 (11.3%)	.76
Male	28 (93.3%)	330 (88.7%)	
Race			
White	28 (93.3%)	350 (94.1%)	.70
Other	2 (6.7%)	22 (5.9%)	
Disease site			
Base of tongue	12 (40.0%)	200 (53.8%)	.18
Tonsil/other	18 (60.0%)	172 (46.2%)	
Smoking			
Current	13 (43.3%)	75 (20.2%)	.005
Former	11(36.7%)	128 (34.4%)	
Never	6 (20.0%)	169 (45.4%)	
Alcohol use			
Current	21 (70.0%)	195 (52.4%)	.05
Former	6 (20.0%)	68 (18.3%)	
Never	3 (10.0%)	109 (29.3%)	
Feeding tube requirement			
Yes	19 (63.3%)	181 (48.7%)	.12
No	11 (36.7%)	191 (51.3%)	
Radiation type			
3D	9 (30.0%)	59 (15.6%)	.07
IMRT	21 (70.0%)	313 (84.1%)	
Radiation dose			
<70 Gy	12 (40.0%)	213 (57.3%)	.07
≥70 Gy	18 (60.0%)	159 (42.7%)	
Treatment y			
2000-2002	9 (30.0%)	73 (19.6%)	.61
2003-2004	8 (26.7%)	113 (30.4%)	
2005-2006	7 (23.3%)	92 (24.7%)	
2007-2008	6 (20.0%)	94 (25.3%)	
T stage			
1	7 (23.3%)	157 (42.2%)	.05
2	23 (76.7%)	215 (57.8%)	
N stage			
0	0	12 (3.2%)	.14
1	4 (13.3%)	56 (15.1%)	
2A	5 (16.7%)	57 (15.3%)	
2B	11 (36.7%)	186 (50.0%)	
2C	9 (30.0%)	42 (11.3%)	
3	1 (3.3%)	19 (5.1%)	
Overall stage			
I	0	1 (0.3%)	1.00
II	0	11 (3.0%)	
III	4 (13.3%)	54 (14.5%)	
IV	26 (86.7%)	306 (82.3%)	
Surgery			
Yes	2 (6.7%)	29 (7.8%)	1.00
No	28 (93.3%)	343 (92.2%)	
Chemotherapy			
Yes	19 (63.3%)	177 (47.6%)	.10
No	11 (36.7%)	195 (52.4%)	

Abbreviations: IMRT = intensity modulated radiation therapy; ORN = osteoradionecrosis; 3D = 3-dimensional.

Table 2 Characteristics of 25 ORN patients and 40 selected ORN-free control individuals

Characteristic	ORN, n (%)	ORN-free, n (%)	*P*
Median age at diagnosis, y (range)	54 (39-80)	53 (40-77)	.81
Sex			
Female	2 (8.0%)	4 (10.0%)	1.00
Male	23 (92.0%)	36 (90.0%)	
Disease site			
Base of tongue	11 (44.0%)	19 (47.5%)	.78
Tonsil/other	14 (56.0%)	21 (52.5%)	
Smoking			
Current	11 (44.0%)	5 (12.5%)	.004
Former	8 (32.0%)	10 (25.0%)	
Never	6 (24.0%)	25 (62.5%)	
Alcohol use			
Current	16 (64.0%)	14 (35.0%)	.03
Former	6 (24.0%)	10 (25.0%)	
Never	3 (12.0%)	16 (40.0%)	
Dental status			
Dentate, without extraction	10 (41.7%)	38 (92.7%)	<.0001
Received dental extraction	9 (37.5%)	0	
Edentulous	5 (20.8%)	3 (7.3%)	
Radiation type			
3D	7 (28.0%)	10 (25.0%)	.78
IMRT	18 (72.0%)	30 (75.0%)	
Radiation dose			
<6996 cGy	10 (40.0%)	20 (50.0%)	.43
≥6996 cGy	15 (60.0%)	20 (50.0%)	
T stage			
1	6 (24.0%)	9 (22.5%)	1.00
2	19 (76.0%)	31 (77.5%)	
Chemotherapy			
Yes	16 (64.0%)	19 (47.5%)	.21
No	9 (36.0 %)	21 (52.5%)	

Abbreviations: IMRT = intensity modulated radiation therapy; ORN = osteoradionecrosis; 3D = 3-dimensional.

However, the difference became nonsignificant ($P=.15$) in the multivariate model after adjustment for age, cancer subsite, sex, treatment type (3D or IMRT), and dental status (dentate or with extraction). V50-V60 were the most significant differences between the ORN group and the comparison group. The differences in V50 were significant in both the univariate (OR = 1.07; 95% CI = 1.02-1.12; $P=.005$) and multivariate (OR = 1.09; 95% CI = 1.01-1.17; $P=.02$) logistic regression models, and the difference in V60 was significant in univariate analysis (OR = 1.04; 95% CI = 1.00-1.08; $P=.04$) and marginally significant in multivariate model (OR = 1.07; 95% CI = 1.00-1.14; $P=.06$). The average of the mean mandible doses was 37.5 Gy for ORN-free patients and 41.5 Gy for ORN patients. The mean mandible dose was associated with ORN in the univariate model but not in the multivariate model, as demonstrated in Table 3.

The locations of ORN were available in 5 of the 6 patients with grade 4 ORN. Review of radiation dose distributions in these 5 patients showed that the locations of ORN were all on the ipsilateral side of the gross disease. The average dose for the regions of ORN was 66 Gy (range, 57-73 Gy), and ORN occurred in or near the areas of the highest prescription dose regions.

Discussion

In this observational study of patients with stage T1-T2 oropharyngeal cancer, ORN developed in 30 of 402 patients (7.5%) during a median follow-up time of 31 months, and 6 (1.5%) eventually needed major surgical intervention. A fairly representative series on the incidence of ORN was reported by Reuther et al; the authors described a 8.2% rate of ORN in 830 patients during a 30-year period (8). This incidence seems consistent with our findings. In general, the incidence of ORN described in the literature varies widely. This is largely due to the differing populations described including the irradiated sites of disease within the head and neck, the cohort size, and the type of radiation delivered (ie, brachytherapy vs teletherapy) (9, 10). Our study focused on patients with small cancers in the oropharynx treated with external beam. These patients always have their posterior mandible in field but receive minimal radiation to the anterior mandible.

The era of radiation is important in a review of the literature, because improvement in techniques (though countered perhaps by more aggressive radiation and chemoradiation strategies) have likely resulted in a decrease in the incidence of ORN. Wahl described a reduction in the incidence of ORN from 11.8% before 1968 to 5.4% from 1968-1992, and again after 1997 to approximately 3% (10). Recently, Lee et al found that the frequency of ORN among 198 patients with either oral cavity or oropharyngeal cancers treated with radiation between 1990 and 2000 was 6.6% (11). Our study was restricted to a 9-year period (2000-2009). The majority of our patients were treated with IMRT, and we did see a trend toward less ORN in our patients treated with IMRT compared with 3D (6% vs 13%).

In addition to radiation techniques, preradiation dental evaluation with prophylactic dental extraction of high-risk teeth in the area of the mandible expected to receive high-dose radiation may lower the rate of ORN. Ben-David et al analyzed 176 head-and-neck cancer patients who received preirradiation dental evaluation and were treated with IMRT (1). Other than dental extractions, their patients also followed a regimen of fluoride gel application and used mouth guards during radiation therapy. No patient experienced ORN during a median follow-up time of 35 months.

Patients' general dental health before radiation therapy is a risk factor for ORN (12), and postradiation dental extraction in the radiation field results in increased ORN risk (12-16). For example, Thorn et al demonstrated that more than half of the ORN cases in their study were initiated by postradiation dental extraction (15). Consequently, it is often considered helpful to extract decaying and nonrestorable teeth before radiation treatment and subsequently maintain good oral hygiene through fluoride gel use and dental surveillance (10). Contrary to common belief, we found that dental extraction before radiation treatment was associated with increased risk for ORN, and none of the 2 ORN patients in this study received preirradiation dental extraction. In addition, a larger proportion of ORN patients were edentulous in the posterior mandible at baseline compared with the matched ORN-free controls. This finding is in line with results from a study by Chang et al, in which 413 patients with oropharyngeal cancer treated with definitive radiation were followed up for a median of 3.8 years (17). Those with poor in-field teeth and preradiation extractions had a higher 5-year incidence of ORN than those who

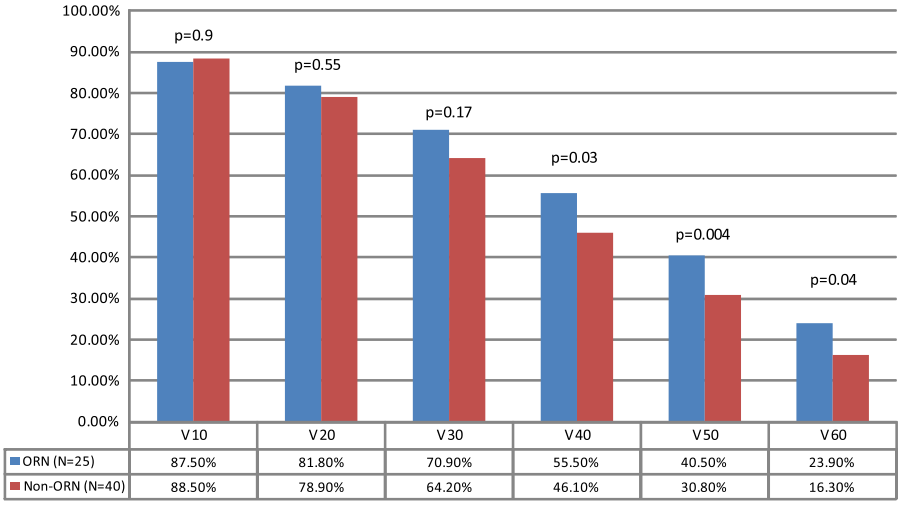

Fig. Mean percent of volume of mandible receiving between 10 and 60 Gy.

did not have preradiation extractions (16% vs 6%), although the difference was not statistically significant ($P=.48$). Unlike our study, in which we observed a larger proportion of ORN patients who were edentulous at baseline compared with the ORN-free controls, Chang et al showed that patients who presented with an edentulous mandible had a <1% risk of ORN, but 14% of patients who were made edentulous from preradiation extractions experienced ORN. However, it is possible that patients requiring dental extractions were more likely to have poor dental hygiene, to smoke, and to have comorbid conditions with poor bone quality/blood supply, thereby having a greater risk of ORN. Another explanation is that the patients who underwent extraction needed a longer time to heal from the procedures, and inadequate healing after any dental extraction could be detrimental and confer an increased risk of ORN.

In the entire study cohort of 402 patients, the majority of patients who experienced ORN were smokers and alcohol drinkers compared with ORN-free patients. These differences remained evident between the ORN and matched ORN-free patients. The observed association between smoking and ORN was demonstrated in a review of 86 head-and-neck cancer patients who participated in a prospective smoking cessation program for current smokers at our institution. Patients who continued to smoke during radiation treatment had a 32% increased risk for the development of ORN and a 46% increased risk for requiring hospitalization during treatment compared with those who quit smoking while undergoing radiation therapy (18). Oh et al have shown that in patients with ORN who continued to smoke or consume alcohol, failure of conservative ORN management and ultimate requirement for surgical resection were more likely (19).

Many studies have evaluated dosimetric parameters and the risk of ORN, and most of them observed a significantly increased risk of ORN in patients receiving high doses of radiation to the mandible. In the study by Chang et al, radiation dose ≥70 Gy was associated with an increased risk of ORN (17). Ben-David et al (1) and Studer et al (5) suggested that ORN may be related to smaller mandibular volumes receiving high doses, because this can exacerbate bony exposure caused by severe acute mucositis. Thus, limiting the mandibular volumes receiving a high dose and reducing hot spots in the mandible can help prevent ORN. Lee et al examined the dose-effect relationship for mandibular ORN in 198 patients with oral cavity and oropharyngeal cancers, and higher radiation doses above biologically equivalent dose values of 102.6 Gy were significantly associated with the development of ORN (11). Gomez et al further demonstrated that a maximum mandibular dose >70 Gy and a mean mandibular dose >40 Gy were associated with increased subsequent dental events and extractions after IMRT (6). No patient experienced ORN during a median follow-up

Table 3 Nested case-control analysis comparing V10-V60 between 25 ORN patients and 40 ORN-free control individuals (reference group)

	Univariate odds ratios (95% confidence intervals)	*P* value	Multivariate conditional odds ratios* (95% confidence intervals)	*P* value
Mean dose (Gy)	1.09 (1.01-1.18)	.04	1.07 (0.95-1.19)	.24
V10	1.00 (0.97-1.03)	.82	0.97 (0.91-1.03)	.31
V20	1.01 (0.98-1.04)	.53	1.00 (0.96-1.05)	.86
V30	1.03 (0.99-1.06)	.12	1.02 (0.98-1.07)	.38
V40	1.05 (1.01-1.10)	.01	1.04 (0.99-1.10)	.15
V50	1.07 (1.02-1.12)	.005	1.09 (1.01-1.17)	.02
V60	1.04 (1.00-1.08)	.04	1.07 (1.00-1.14)	.06

Abbreviation: ORN = osteoradionecrosis.

* Adjusting for age, cancer subsite, sex, treatment type (3-dimensional or intensity modulated radiation therapy), and dental status (dentate or with extraction).

time of 35 months. Their dose constraint for IMRT included a maximal mandibular dose of <72 Gy and a mean parotid gland dose of ≤26 Gy. Similarly, Gomez et al retrospectively reviewed 168 patients with cancers of the oral cavity, nasopharynx, larynx/hypopharynx, sinus, and oropharynx treated with IMRT. Only 2 (0.7%) patients experienced ORN that required sequestrectomy without major surgery during a 37-month period (6).

In our series there was a trend for the use of IMRT to be associated with a lower incidence of ORN compared with 3-D conformal radiation therapy ($P = .07$). Our practice set dose constraints on the parotid gland similar to that described by Gomez et al (6). With regard to the mandible, our constraint was to attempt to keep the maximal dose to the mandible at or below the prescription dose to the adjacent target volume to minimize hot spots in the mandible. We did not routinely use mean dose objectives to the mandible as part of our treatment prescription.

In this study, we found significant differences in the V40-V60 range between the cases and controls. Even after adjustment for the matching variables and other possible confounders, the differences in V50 and V60 between the 2 groups of patients were still evident, although the *P* value of V60 became marginally significant. Moving forward with IMRT, incorporating mean dose constraints on the mandible in addition to maximal dose constraints should further help lower the rate of ORN.

This study has some limitations. As a retrospective study, the observed associations could be explained by unmeasured confounding variables. In addition, the incidence of postradiation xerostomia was not measured for this study, and radiation exposure of salivary glands can have an impact on the risk of xerostomia and ORN (3, 20). Furthermore, radiation dose to the area of the mandible affected by ORN was unavailable for patients with grades 1-3 ORN; thus, we could not know if a higher dose (or hot spot) delivered to a given location of the mandible corresponded to ORN in that location.

Because the aim of this study was to focus on the effect of radiation on ORN in patients treated with definitive (chemo) radiation only, we decided to include only oropharyngeal cancer patients and to exclude patients with oral cavity cancers, who are most often treated postoperatively and to lower doses. Regardless, this is one of the largest studies to evaluate the association between dosimetric variables and ORN in oropharyngeal cancer patients treated with definitive radiation. We also used a nested case-control design and multivariate models to control for demographic and treatment-related variables. Finally, the study included only patients with stage T1 and T2 lesions. Because this study's patient population was derived from a database used in a study of biomarkers with only limited numbers of stage T3 and T4 disease, we thought that it was adequate to exclude stage T3 or T4 patients from the study to create a more homogenous population.

Conclusion

The overall ORN occurrence was relatively low among oropharyngeal cancer patients receiving radiation therapy, and most ORN patients could be treated conservatively. ORN was associated with the use of tobacco and alcohol and occurred more frequently in patients receiving higher prescription doses (≥70 Gy) to the primary cancer site. The dose-volume factor for the mandible seemed to play a significant role: the range between V50 and V60 saw the most significant differences between the ORN group and the comparison group, suggesting that minimizing the percent mandibular volume exposed to 50 Gy could reduce ORN risk.

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